$$W(v) = \frac{\prod_{i<j}(v_i - v_j)}{1!\ 2!\ 3!\ 4!}$$

$$W(v) = \det\left[\binom{v_i}{j}\right]_{i,j=0}^{4} \in \mathbb{Z}$$

Distinct residues modulo 5 make every factor nonzero

$$W(v) \equiv 1 \pmod 5$$